

## Notes: Factoring Out the Greatest Common Factor

Find the greatest common factor.

1. 9, 27, 33

2.  $12x$ ,  $6x^3$ ,  $24x^2$

3. 14, 28, 56

4.  $11x^2$ ,  $22x^3$ ,  $33x^2$

5.  $32x^2$ ,  $72x^3$ ,  $144x^2$

6.  $3y^2$ ,  $6y^2$ ,  $22y$

7. 82,  $14c$ ,  $22c^2$

Find the missing GCF to complete the factorization.

8.  $3x^2 - 6x = \underline{\hspace{2cm}}(x - 2)$

9.  $a^4 + 7a^2 = \underline{\hspace{2cm}}(a^2 + 7)$

10.  $5y^2 + 25y^3 = \underline{\hspace{2cm}}(1 + 5y)$

11.  $4x^5 - 8x^4 = \underline{\hspace{2cm}}(x - 2)$

Factor the expression.

8.  $6x^2 + 12x$

9.  $9y^2 - 21y$

10.  $8a - 24a^2 + 24a^3$

11.  $q^2 + 16q$

12.  $4k^2 - 20k - 12k^3$

13.  $12h^2 - 36h$

14.  $13x^2 - 26x$

15.  $14d^2 - 35d$

16.  $16x^3 - 4x^4$

17. The area of a rectangle is  $2x^2 - 36x$ . The length of one side is  $2x$ . Find the length of the other side.

18. The width of a rectangle is  $8x^2$  and the area is  $24x^3 + 16x^2$ .

a. What is the length?

b. What is the perimeter?

19. If the area of a rectangle is  $5y - 25y^2$ , and the length is  $5y$  . . .

a. What is the width ?

b. What is the perimeter?